



Berkeley
ENGINEERING

Ackermann Steering Mechanism

ME 130 Design of Planar Machinery

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1 Introduction

This project aims to apply the principles and techniques learned in the course Design of Planar Machinery to the synthesis and analysis of the Ackermann steering mechanism. This study explores key concepts of planar mechanism design including toggle points, transmission angles, and instant centers.

2 Constraints

2.1 Initial Synthesis Constraints

The steering mechanism is constrained in four key ways. First, the extended lines of links R2 and R3 must intersect at the center of the rear axle (see appendix A), which defines the relative positions and angles of R2 and R4. Second, steering angle relationships constrain the linkage to produce correct wheel angles across the full turning range, especially during tight turns. Third, packaging and mounting space constrain the lengths and positions of the links to fit within the vehicle chassis and avoid interference with other components. Fourth, to ensure symmetric steering, R4 should mirror R2 so that the vehicle's turning radius is consistent in both directions.

2.2 Self-Imposed Constraints

To optimize performance, the mechanism should be designed for effective force transmission and to avoid toggle conditions. The transmission angle should remain above 40° (ideally as close to 90° as possible) and the toggle points must be positioned to avoid limiting the turning radius of the vehicle.

3 Design Methods

The mechanism was designed as a four-bar linkage steering system to mimic the Ackermann configuration using rigid links. This topology was selected for its geometric simplicity and mechanical reliability. The main goal was to create a symmetrical linkage where extended steering arms would ideally intersect at the midpoint of the rear axle, supporting pure rolling of the front wheels during turning.

The synthesis began by defining fixed vehicle parameters: wheelbase, track width, and pivot spacing. The ground link (R1) represented the distance between the steering pivots, and the input (R2) and output (R4) links were assumed to be symmetric to enable equal turning radii in both directions. The coupler (R3) was defined based on the known angles and lengths of the other links.

To perform position synthesis, the loop equations were solved numerically using MATLAB to compute the corresponding link orientations. This allowed for parametric exploration of the linkage as a function of R2 length.

After generating the kinematic solution, functional analysis focused on identifying toggle points and evaluating transmission angle performance. The toggle positions—where the linkage approaches a straight-line configuration and loses mechanical advantage—were computed to ensure they did not limit the range of motion. Transmission angle was used as a proxy for force transfer efficiency; the design was refined to optimize the transmission angle across the steering range.

To guide design iterations, key plots were generated including minimum transmission angle vs. linkage dimensions and toggle point location vs. linkage dimensions. These were used to evaluate how different R2 and R3 lengths affected mechanism feasibility and effectiveness. The design was tuned for sufficient packaging space and motion without mechanical interference.

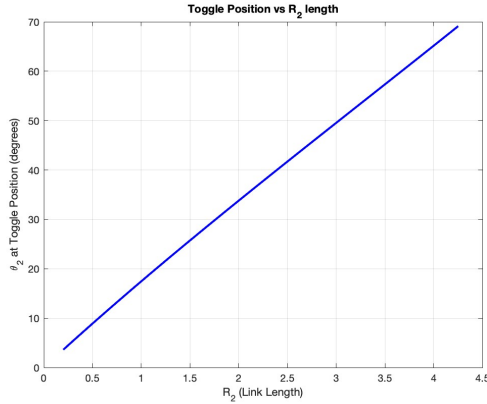
This approach enabled rapid design iteration while meeting physical constraints and ensuring smooth motion over the intended range of steering angles.

4 Results

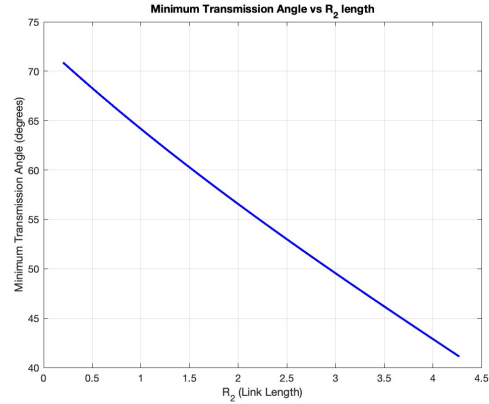
The results of the analysis are divided into three primary components: position analysis, toggle point analysis, and transmission angle evaluation. See Appendix B for angle conventions. The position analysis was formulated so that all relevant angles and link positions were automatically computed based on the chosen length of R2. The neutral configuration of the mechanism was then plotted for visualization.

For the toggle point analysis, the linkage was evaluated over a sweep of R_2 lengths ranging from 0.2 inches to the maximum possible value (just before R_3 becomes zero). At each step, the turning range was analyzed to determine where the configuration approached a toggle position, defined as when the linkage became nearly collinear and lost mechanical advantage. The resulting toggle regions are shown in Figure 1a.

Following this, the minimum transmission angle was calculated in the same R_2 range. The transmission angle serves as an indicator of mechanical efficiency, with angles below 40° suggesting poor force transmission. The plot in Figure 1b shows that as R_2 increases, the minimum transmission angle drops significantly. Although the minimum transmission angle remained above 40° for all R_2 lengths, it is preferable to keep it as close to 90° as possible for optimal force transmission.



(a) Toggle Position Analysis



(b) Transmission Angle Analysis

Figure 1: Analysis results

From this analysis, it is evident that a shorter R_2 link improves both toggle performance and transmission angle. As R_2 increases, the linkage becomes more prone to locking and less mechanically efficient. Based on this, the final prototype was constructed with the following dimensions (see Figure 2): $R_1 = 3.00''$, $R_2 = 0.50''$, $R_3 = 2.65''$, $R_4 = 0.50''$. It is important to note that these results hold if all dimensions of the car are scaled equivalently.



Figure 2: Physical prototype of Ackermann steering mechanism.

4.1 Discussion

The general motivations of the project were achieved. By applying concepts such as transmission angle, instant centers, and toggle points, I was able to better understand the function of the Ackermann steering mechanism and deepen my understanding of planar machinery by applying theoretical concepts to a practical problem.

A Graphical Synthesis Process

The graphical synthesis for the Ackermann steering mechanism is very straightforward. It is easiest to synthesize the mechanism in its neutral position, where neutral is defined as the position where the mechanism is symmetric. The angles of links R2 and R4 are determined by drawing two lines from the pivot points on the ground through the center of the rear axle.

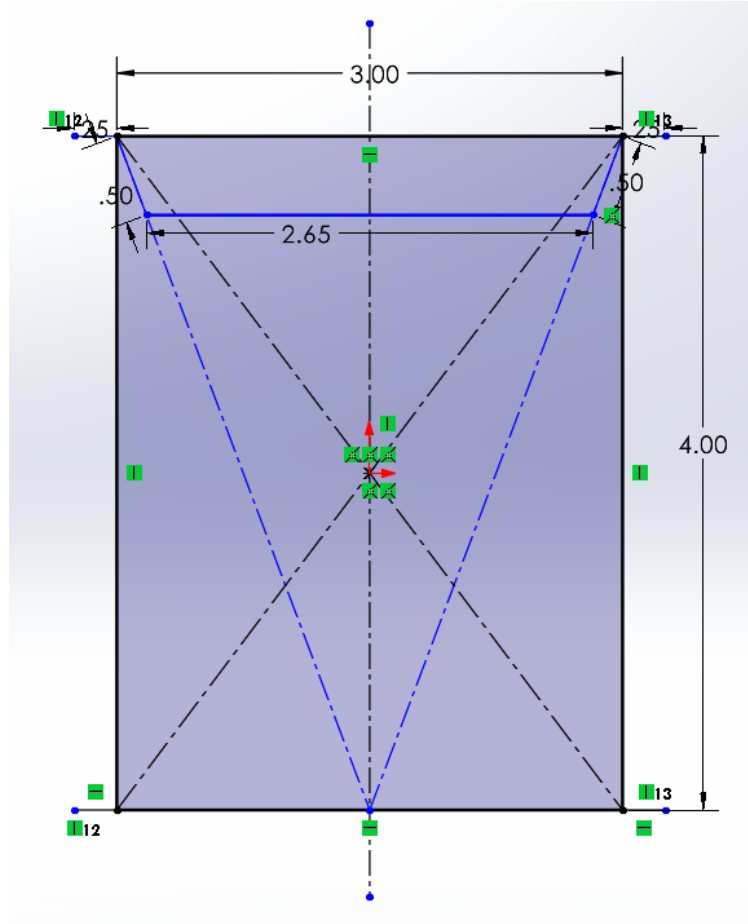


Figure 3: Graphical synthesis of Ackermann steering mechanism.

B MATLAB Code

See the following pages for the MATLAB code used in the analysis. Figure 4 illustrates the angle conventions applied to the linkage. Note that while the magnitude of θ_2 is used in the plot shown in Figure 1a for clarity, the actual value of θ_2 is negative throughout the range of motion, consistent with the standard counterclockwise-positive convention.

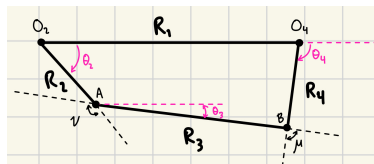


Figure 4: Angle convention used for the matlab code.

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- [Position Synthesis](#)
- [Evaluating Toggle Points for different link lengths](#)
- [Evaluating minimum transmission angle for different link lengths](#)

Position Synthesis

```
clc;
clear all;

% Define Vehicle Parameters and angle over which the wheels turn
wt = 3.5;          % Wheel Track Length
wb = 4;            % Wheel Base Length
pivot_dist = 3;    % Distance between ground pivots
rat = wb / wt;
thetain = -95:0.1:-20;
thetain = thetain*(pi/180);

beta = atand((pivot_dist/2)/wb);
r2_max = ((pivot_dist/2)^2 + wb^2)^0.5;

% Define angles
theta_1 = 0;
theta_2_init = -(90 - beta);
theta_2_rad_init = deg2rad(theta_2_init);

% Define all link lengths/ magnitudes parametrically
c = 0.117;
r1 = pivot_dist;
r2 = c * r2_max;
r4 = r2;
r3 = r1 - 2*r2*cosd(theta_2_init);

% Form Loop Equation
% R2 + R3 = R4 + R1;
% r2*exp(i*theta_2_rad) + r3*exp(i*theta_3) == r4*exp(i*theta_4) + r1*exp(i*theta_1);

syms theta_3 theta_4

eqn1 = real(r2*exp(1i*theta_2_rad_init) + r3*exp(1i*theta_3) - r4*exp(1i*theta_4) - r1*exp(1i*theta_1)) == 0;
eqn2 = imag(r2*exp(1i*theta_2_rad_init) + r3*exp(1i*theta_3) - r4*exp(1i*theta_4) - r1*exp(1i*theta_1)) == 0;

sol = vpasolve([eqn1, eqn2], [theta_3, theta_4]);
theta_3 = double(rad2deg(sol.theta_3));
theta_4 = double(rad2deg(sol.theta_4));
theta_3_rad = rad2deg(theta_3);
theta_4_rad = rad2deg(theta_4);

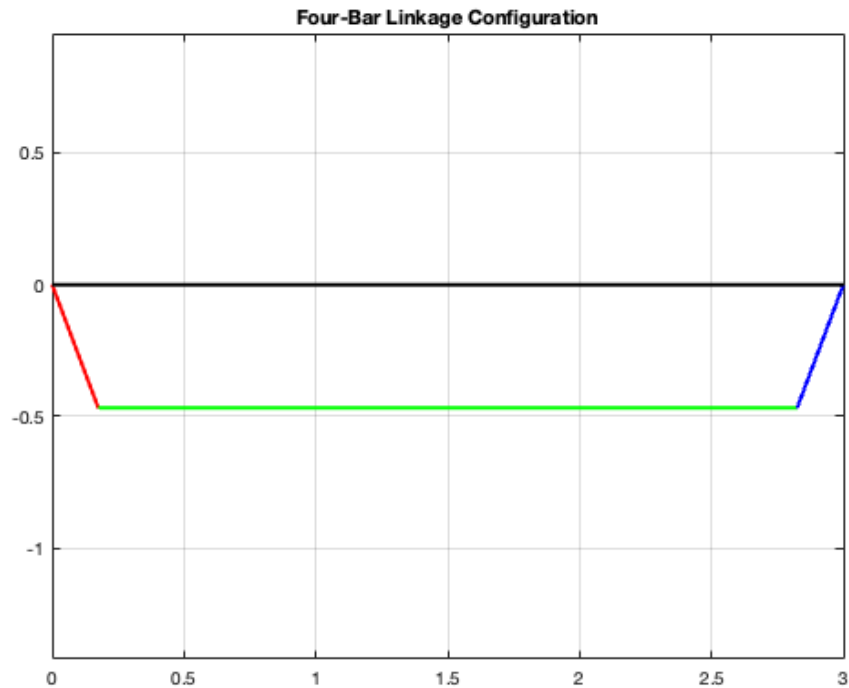
% Calculate Toggle Points
% Toggle between R2 and R3
theta_2_min = acosd((r1^2 + (r2+r3)^2 - r4^2) / (2*r1*(r2+r3)));
% Toggle between R3 and R4
theta_4_min = 180 - (acosd((r1^2 + (r3+r4)^2 - r2^2) / (2*r1*(r3+r4))));
theta_2_max = acosd((r1^2 + r2^2 - (r3+r4)^2) / (2*r1*r2));

% Plot
O2 = [0, 0];
A = O2 + r2 * [cos(theta_2_rad_init), sin(theta_2_rad_init)];
B = A + r3 * [cosd(theta_3), sind(theta_3)];
O4 = [r1, 0];
```

```

figure;
plot([O2(1), A(1)], [O2(2), A(2)], 'r-', 'LineWidth', 2); hold on;
plot([A(1), B(1)], [A(2), B(2)], 'g-', 'LineWidth', 2);
plot([B(1), O4(1)], [B(2), O4(2)], 'b-', 'LineWidth', 2);
plot([O4(1), O2(1)], [O4(2), O2(2)], 'k-', 'LineWidth', 2);
axis equal;
grid on;
title('Four-Bar Linkage Configuration');

```



Evaluating Toggle Points for different link lengths

```

r2_vals = linspace(0.2, r2_max, 200);
theta2_min_vals = NaN(size(r2_vals));

for i = 1:length(r2_vals)
    r2 = r2_vals(i);
    r4 = r2; % Assuming symmetric (like Ackermann linkage)

    % Calculate r3 based on geometry
    r3 = r1 - 2*r2*cosd(theta2_init);

    % Form triangle: r4, r2+r3, r1 (ground)
    a = r1;
    b = r2 + r3;
    c = r4;

    % Check for valid triangle
    if r3 + r2 > r4 && r1 < (r3 + r2 + r4)
        % Law of cosines to find theta2_min
        cos_theta2_min = (a^2 + b^2 - c^2) / (2 * a * b);

        % Check if in valid range
        if abs(cos_theta2_min) <= 1
            theta2_min_vals(i) = acosd((a^2 + b^2 - c^2) / (2 * a * b));
        end
    end
end

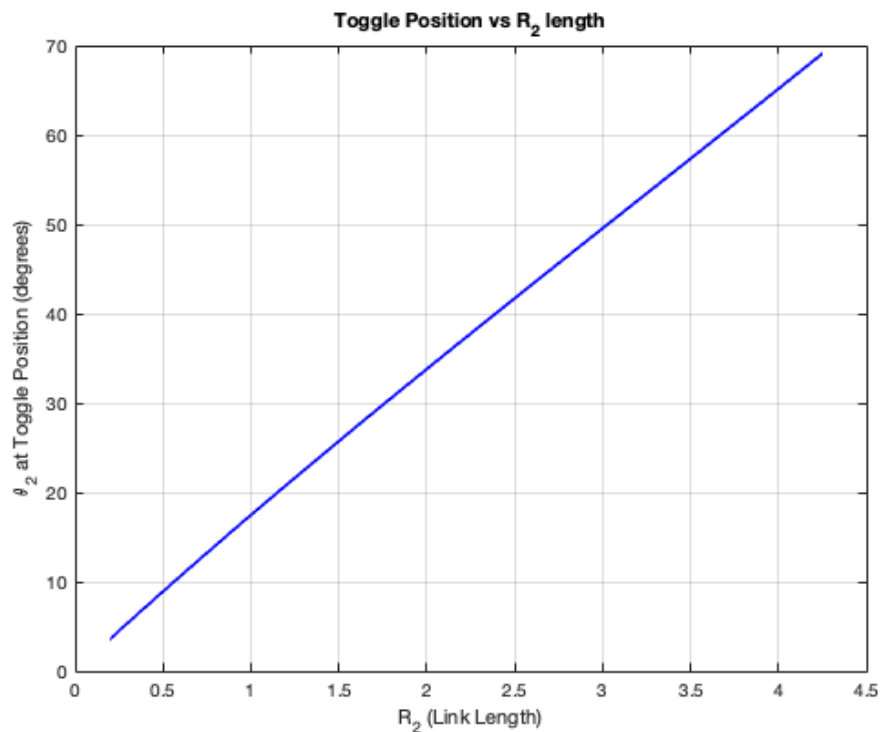
```

```

        end
    end
end

% Plot result
figure;
plot(r2_vals, theta2_min_vals, 'b-', 'LineWidth', 2);
xlabel('R_2 (Link Length)');
ylabel('\theta_2 at Toggle Position (degrees)');
title('Toggle Position vs R_2 length');
grid on;

```



Evaluating minimum transmission angle for different link lengths

```

r2_vals = linspace(0.2, r2_max, 200);
min_trans_angle_vals = NaN(size(r2_vals));

for i = 1:length(r2_vals)
    r2 = r2_vals(i);
    r4 = r2;

    % Calculate r3 based on geometry
    r3 = r1 - 2*r2*cosd(theta_2_init);

    % Form triangle: r4, r2+r3, r1 (ground)
    a = r4;
    b = r2 + r3;
    c = r1;

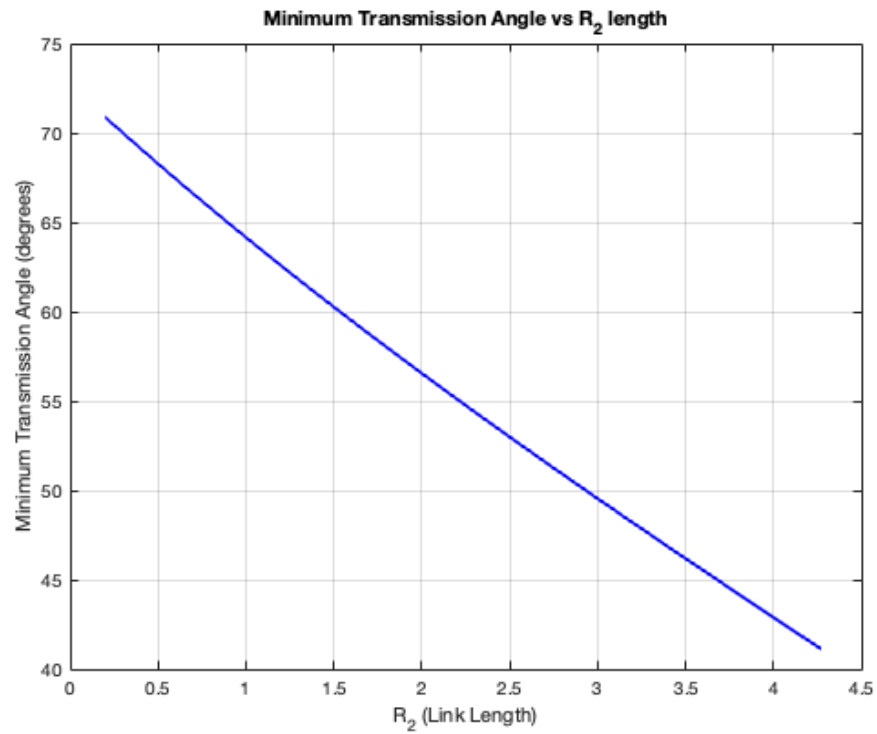
    % Check for valid triangle
    if a + b > c && abs(a - b) < c
        % Law of cosines to find angle between r4 and (r2 + r3)
        cos_mu = (a^2 + b^2 - c^2) / (2 * a * b);
        cos_mu = min(max(cos_mu, -1), 1); % Cut off to avoid NaN
        mu = acosd(cos_mu);
        min_trans_angle_vals(i) = mu;
    end
end

```



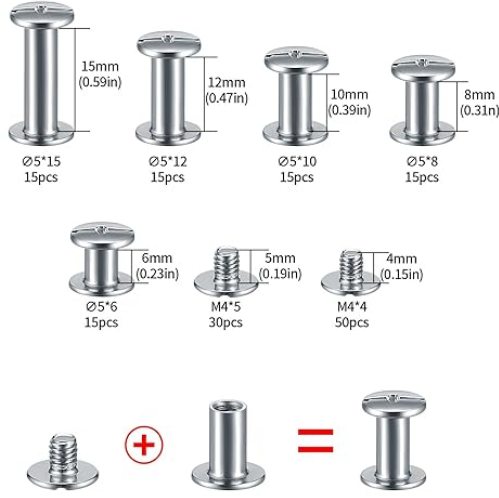
```
end
end

% Plot result
figure;
plot(r2_vals, min_trans_angle_vals, 'b-', 'LineWidth', 2);
xlabel('R_2 (Link Length)');
ylabel('Minimum Transmission Angle (degrees)');
title('Minimum Transmission Angle vs R_2 length');
grid on;
```



C Prototype Fabrication Details

The prototype was fabricated for less than five dollars. All linkage components were 3D printed using PLA filament. Pin joints were implemented using binding screws and posts with plastic washers between components. The binding screws did become loose after many rotations, but they worked well enough without any thread locker for demonstration purposes.



(a) Binding Screws

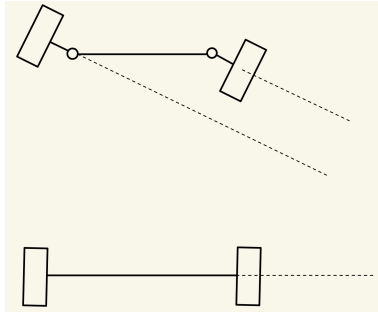


(b) Plastic Washers

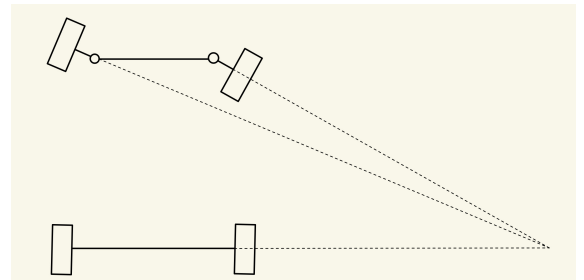
Figure 5: Pin Joint Materials

D Additional Information

The Ackermann steering mechanism is designed to approximate ideal kinematic conditions for planar vehicle turning. Specifically, it ensures that the extended axes of the front wheels intersect at a common point, ideally located along the rear axle, which serves as the instantaneous center of rotation for the vehicle during low-speed turns. This configuration minimizes lateral slip at the front wheels and reduces tire scrub, unlike parallel steering, where the wheels follow distinct turning paths and cannot maintain pure rolling. Ackermann geometry achieves this by setting the inner wheel at a greater steering angle than the outer, matching the difference in turning radius. A visual comparison between Ackermann and parallel steering is shown in Figure 5.



(a) Parallel Steering



(b) Ackermann Steering

Figure 6: Comparison of Ackermann Steering and Parallel Steering